

MONITOR & RESPOND TO NETWORK SECURITY EVENTS

Network Security Assignment #7 | DeRexi: Policy Pilot

SCENARIO STATUS FICTIONAL / EDUCATIONAL. This report adapts the prior DeRexi APT38 / BeagleBoyz case study for the Network Security monitoring-and-response rubric. The incident timeline and log entries are simulated evidence, not production records.

01 // INCIDENT SUMMARY

Security monitoring identifies a suspected DeRexi compromise

In this fictional scenario, a DeRexi employee receives a client-themed spearphishing attachment. After the file is opened, monitoring begins to show unusual PowerShell execution, creation of a previously unseen Windows service, repeated authentication attempts, directory and network-share enumeration, suspicious data staging, and later file-deletion / timestamp anomalies.

At 10:09, the Security Operations Center (SOC) correlates the endpoint, identity and discovery alerts and declares a suspected APT38 / BeagleBoyz intrusion. Because DeRexi supports financial institutions, the event is treated as a high-severity potential supply-chain incident until forensic review is complete.

SECURITY INCIDENT IDENTIFIED Suspected APT38 / BeagleBoyz compromise of a DeRexi employee endpoint with potential access to internal policy resources and administrative pathways.

02 // SIMULATED INCIDENT LOG

From alert activity to containment

Time	Source / Event	Monitoring evidence	Analyst action
08:47	Endpoint - PowerShell	Unusual PowerShell execution after attachment opened.	Open investigation; preserve endpoint telemetry.
09:03	Endpoint - Service creation	Previously unseen Windows service appears.	Escalate persistence suspicion.
09:18	Identity - Authentication	Repeated access attempts against additional internal accounts.	Review account activity; prepare credential containment.
09:31	Network / endpoint - Discovery	Directory and network-share enumeration begins.	Correlate activity with policy-resource access.
10:02	Endpoint - Defense evasion	Temporary artifacts disappear; timestamp anomalies observed.	Preserve logs and increase incident severity.
10:09	SOC correlation	PowerShell + service creation + discovery alerts align.	Declare suspected compromise.
10:16	IR response	Endpoint isolated; accounts disabled; sessions revoked.	Contain affected access paths.

03 // RESPONSE WORKFLOW

How DeRexi contains and responds

1 CONTAIN	Isolate affected endpoints, disable suspected accounts, revoke active sessions and restrict administrative / client-facing access paths.
2 PRESERVE EVIDENCE	Centralize endpoint, identity, application and cloud logs so attacker deletion on one host does not erase the investigation trail.
3 VALIDATE INTEGRITY	Compare policy documents and versions against trusted hashes / version history to identify unauthorized policy changes.
4 PROTECT PRIVILEGED ACCESS	Require phishing-resistant MFA and separate administrative identities from ordinary user accounts.
5 DETECT RELATED BEHAVIOR	Alert on suspicious PowerShell, new services, repeated authentication attempts, unusual share enumeration, file deletion and timestamp anomalies.
6 SEGMENT CLIENTS	Maintain tenant and administrative boundaries so compromise of one employee or component cannot become unrestricted access to banking clients.
7 ESCALATE & COMMUNICATE	Activate incident response, leadership, legal/compliance and banking-client communication procedures when regulated customers could be affected.

04 // SCREENSHOT / LOG EVIDENCE

Simulated SOC monitoring snapshot

SIMULATED SOC LOG SNAPSHOT - EDUCATIONAL EVIDENCE

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08:47:12 EDR HIGH powershell.exe spawned after document open
09:03:41 EDR HIGH new Windows service created: unknown baseline
09:18:07 IAM MEDIUM repeated authentication attempts across internal accounts
09:31:26 NDR MEDIUM SMB/share enumeration from employee endpoint
09:46:55 EDR MEDIUM local files accessed and staged in temporary path
10:02:14 EDR HIGH artifact deletion + timestamp anomaly detected
10:09:03 SOC CRITICAL correlated alerts -> suspected compromise declared
10:16:22 IR ACTION endpoint isolated | sessions revoked | account disabled

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EVIDENCE NOTE The log snapshot above is a simulated visual created from the incident timeline already documented in the DeRexi APT38 case study. It is included to demonstrate how monitoring evidence could be presented for this educational assignment; it is not a screenshot from a live production SOC.

05 // OUTCOME & CONCLUSION

For this case study, DeRexi contains the intrusion before investigators find evidence of a successful compromise of a client banking environment. The incident is nevertheless treated as a potential supply-chain event until forensic review is complete. The monitoring process demonstrates how endpoint, identity and network signals become more meaningful when correlated, and the response sequence shows how detection must lead quickly to containment, evidence preservation, access control and recovery actions.

Academic disclosure: all events involving DeRexi are fictional and were created for the assignment. The APT38 / BeagleBoyz behaviors used to structure the scenario come from the previously prepared MITRE ATT&CK / CISA-based case study.